

Problem

On the Bode phase plot for $0.5 < \omega < 50$, the slope of $[1 + j10\omega - \omega^2/25]^2$ is

- (a) $45^\circ/\text{decade}$
- (b) $90^\circ/\text{decade}$
- (c) $135^\circ/\text{decade}$
- (d) $180^\circ/\text{decade}$

Step-by-step solution

Step 1 of 3

For the factors in form of $\left[1 + \frac{2j\omega\xi}{\omega_n} + \left(\frac{j\omega}{\omega_n}\right)^2\right]$, the slope of the magnitude plot is

$40N$ dB/decade and that of the phase plot is $180N^\circ$ per decade.

Apply the phenomenon to the factor $\left[1 + j10\omega - \frac{\omega^2}{25}\right]^2$, and then slope of the magnitude plot is

$40(2)$ dB = 80 dB and that of the phase plot is $180(2)^\circ = 360^\circ$ due to power of N .

Step 2 of 3

Draw the phase plot for factor $\left[1 + j10\omega - \frac{\omega^2}{25}\right]^2$.

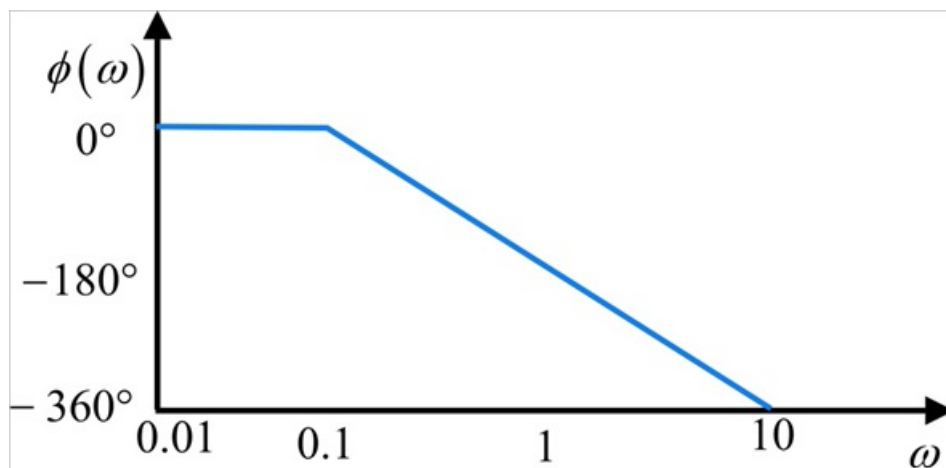


Figure 1

Step 3 of 3

From the figure 1 it is clear that the slope of the factor phase is $180^\circ/\text{decade}$.

Therefore On the bode phase plot for $0.5 < \omega < 50$, the slope of $\left[1 + j10\omega - \frac{\omega^2}{25}\right]^2$ is

$180^\circ/\text{decade}$. Thus the correct option is **(d)**. So remaining options (a), (b) and (c) are wrong.